

Research Notes on Artificial neural network

Artificial neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

>> Section 1

The learning rate defines the size of the corrective steps that the model takes to adjust for errors in each observation. A high learning rate shortens training time, but with lower ultimate accuracy, while a lower learning rate takes longer, but with the potential for greater accuracy. Optimizations such as quickprop are primarily aimed at accelerating learning. In order to avoid oscillations such as connection weights that cycle between high and low values, and to improve the rate of convergence, refinements use an adaptive learning rate that increases or decreases as appropriate. The concept of momentum allows the balance between the gradient and the previous change to be weighted such that the weight adjustment depends to some degree on the previous change. A momentum close to 0 emphasizes the gradient, while a value close to 1 emphasizes the last change.

>> Section 2

Image processing NNs are employed in computer vision tasks such as image classification, object and facial recognition, and image segmentation. They have been applied to automated surveillance medical imaging for diagnosis.

>> Section 3

Applications Neural networks support a broad range of applications in image processing, speech recognition, natural language processing, finance, and medicine. Because of their ability to model and reproduce nonlinear processes, neural networks have found applications in many disciplines. These include:

>> Section 4

Medicine NNs analyze medical datasets. They enhance diagnostic accuracy, especially by interpreting complex medical imaging for early disease detection, and by predicting patient outcomes for personalized treatment planning. In drug discovery, NNs speed up the identification of potential drug candidates and predict their efficacy and safety, significantly reducing development time and costs. Additionally, their application in personalized medicine and healthcare data analysis allows tailored therapies and efficient patient care management.

>> Section 5

Capacity A model's "capacity" property is its ability to model any given function. It is related to the amount of information that can be stored in the network and to the notion of complexity. Information capacity and the VC dimension are two metrics. The capacity of a network of standard neurons (not convolutional) can be derived by four rules that derive from considering a neuron as an electrical element. The information capacity captures the functions that the network can model given any data as input. The VC dimension uses the principles of measure theory and finds the maximum capacity under optimal circumstances, given input data in a specific form. The VC Dimension for arbitrary inputs is half the information capacity of a perceptron. The VC Dimension for arbitrary points is sometimes referred to as Memory Capacity.

>> Section 6

NNs are composed of digital neurons, conceptually derived from biological neurons. Each neuron has one or more numerical inputs and produces a single numerical output. An input (e.g., an image) is typically parceled across the set of input neurons (each getting a piece of the image). Each neuron's inputs, weighted by the weights of the connections from each of those inputs are summed. A bias term is added to this sum. bias) The result is then passed through a nonlinear activation function to produce the neuron's output. Small enough outputs may be zeroed out (ignored).

>> Section 7

Backpropagation is a method used to adjust the connection weights to compensate for errors found during learning. The error amount is essentially divided among the connections. Technically, backpropagation calculates the gradient (the derivative) of the loss function associated with a given state with respect to the weights. The weight updates can be done via stochastic gradient descent or other methods, such as extreme learning machines, "no-prop" networks, training without backtracking, "weightless" networks, and non-connectionist neural networks.

>> Section 8

Training NNs require millions of training samples to become functional. As of 2026, training a commercial LLM (GPT, Grok, Gemini) typically required hundreds of thousands of computers, and cost 10s of millions of dollars.

>> Section 9

Dataset bias Neural networks are dependent on the quality of their training data; low quality data can lead the model to product poor results. Biased data produces biased results,

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requiring trainers to detect and correct them. For example, data that underrepresents some demographic groups may prevent cause errors, e.g., in facial recognition and law enforcement. In 2018, Amazon scrapped a recruiting tool because the model favored men over women for jobs in software engineering due to the larger number of male workers in the field. The system penalized resumes with the word "woman" or the name of a women's college. One corrective is to add synthetic data to offset the bias.

>> Section 10

Computational power The multilayer perceptron is a universal function approximator, as proven by the universal approximation theorem. However, the proof does not specify the number of neurons required, the network topology, the weights or the learning parameters. A recurrent architecture with rational-valued weights (as opposed to full precision real number-valued weights) has the power of a universal Turing machine, using a finite number of neurons and linear connections. Further, the use of irrational values for weights results in a machine with super-Turing power.

>> Section 11

Applications that must generalize well to unseen examples, must avoid over-training. This arises in convoluted or over-specified systems when the network capacity is much larger than needed. Two approaches address over-training. Cross-validation and similar techniques can check for over-training and select appropriate hyperparameters to minimize generalization error. Regularization in a probabilistic (Bayesian) framework can be performed by selecting a larger prior probability over simpler models; but also in statistical learning theory, where the goal is to minimize two quantities: the 'empirical risk' and the 'structural risk', which roughly corresponds to the error over the training set and the predicted error in unseen data due to overfitting.

>> Section 12

Training/learning involves adjusting the weights of the network to improve the accuracy of the result. NNs typically require vast numbers of sample inputs (far more than biological brains) to achieve a given level of function. This is done by minimizing the observed errors among sample observations. Training takes place before a network is deployed, and (unlike brains) does not continue thereafter. Instead, the network may be retrained from scratch as more sample data becomes available. Empirical risk minimization adjusts node and link weights to minimize the difference (empirical risk), between the predicted output and the known values in the training samples. A defined loss function measures the degree of error.

Backpropagation spreads the error (adjusts the weights) from the output nodes across the network to the input nodes. The intent is to allow the network to process data that is not included in the training samples. As long as the value of the loss function (its cost) continues to decline, the network is continuing to improve. The function typically produces a statistic whose value is only approximate. When the cost is low, the difference between the output (e.g. almost certainly a cat) and the correct answer (a cat) is small. Most learning models can be viewed as a straightforward application of optimization theory and statistical estimation. Learning typically ends when additional observations do not usefully reduce the cost. The cost typically approaches, but does not reach, 0. If no feasible amount of sample data yields a low cost, training is labeled a failure.